

Finite but Boundaryless

A WRRRA Hybrid Expansion and Accumulated Twist Cosmology with Hidden
Gravitational State

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Abstract

This paper formulates a new upstream cosmological hypothesis within the Wonsik Reality Renderer Architecture. The starting intuition is that physical reality should not require an actually infinite stored state. A finite universe need not possess an edge, however, and finiteness alone does not select a unique topology or produce gravity. The model therefore makes a more specific and falsifiable choice: a distributed local twist can accumulate along transport and leave a nontrivial global holonomy whose local residues enter gravitational dynamics even when the corresponding information is not rendered as visible matter. A Möbius strip supplies only the simplest visual example of a twist distributed continuously around a closed form. It is not a single physical cut or seam and is not proposed as the literal topology of the universe because it has a boundary and is two dimensional.

The paper separates three claims that are often conflated. Compactness concerns global extent, curvature concerns local geometry, and holonomy concerns the residue obtained by transporting a state around a closed path. A flat three torus is compact and boundaryless while remaining locally flat, so topology alone cannot explain excess gravity. WRRR requires an explicit causal chain from global identification to holonomy, from holonomy to a local twist residue, and from that residue to an effective stress tensor. The minimal bookkeeping target is

$$C \rightarrow A_T \rightarrow H \rightarrow \Theta \rightarrow (\rho_T, q_T) \rightarrow T_{\mu\nu}^T \rightarrow O.$$

Cosmic expansion and twist evolution are treated as independent state variables. The spatial metric is parameterized as $g_{ij} = a^2 \tilde{g}_{ij}(\Theta)$, allowing expansion only, twist only, and hybrid histories. The exact photon redshift remains $1 + z = (u_\mu k^\mu)_e / (u_\mu k^\mu)_o$. A proposed factorization, $1 + z_{obs} = (a_0 / a_e) e^{\Xi_T}$, is a conditional hypothesis rather than a derived result. It becomes scientific only when Ξ_T is frozen by the same twist state that also determines galactic additional gravity, disk plane geometry, warps, vertical equilibrium, and lensing.

At galactic scale the proposal connects to an existing WRRR Milky Way model in which the unoriented director $[n]$ selects a local disk plane but does not define angular momentum. The magnitude and anisotropic sectors (ρ_T, q_T) generate a candidate dark mass phenotype. The present work does not provide a complete covariant action or a cosmological likelihood. It provides a disciplined model space, an ownership map for every equation, and a preregistered falsification program spanning supernova time dilation, distance duality, Tolman dimming, BAO, the CMB, redshift drift, structure growth, lensing, galaxy dynamics, vertical equilibrium, and Solar System limits.

Keywords WRRR, finite universe, boundaryless universe, cosmic topology, holonomy, twist stress, expansion, cosmological redshift, dark matter phenotype, galactic warp

1 Research Question and Main Conclusion

The research question is not simply whether a finite universe is mathematically possible. Compact boundaryless spatial manifolds are already standard mathematical possibilities. The question is whether a finite globally closed state can leave a dynamically reusable residue that explains more than one observational sector.

The main conclusion is conditional. A global twist is not implied by finiteness, and gravitational stress is not implied by topology. A viable WRRRA theory must add a dynamical map from global transport to local stress and must freeze that map before confronting multiple data sets. If the same state Θ accounts for a component of cosmological redshift and for the additional gravitational and geometric outputs of galaxies, the proposal is more than a relabeling of expansion or dark matter. If separate arbitrary functions are introduced for each observation, the proposal collapses into an unfalsifiable reparameterization.

This paper therefore presents a structural hypothesis, not a completed replacement for general relativity or the standard cosmological model. Its central object is accumulated twist, not a one time topological flip at a privileged section.

1.1 Conceptual Origin

The first image of accumulated spatial twist in this work arose while viewing **KOO JEONG A: OUSSMOS** at Leeum Museum of Art in Seoul. The exhibition was presented in M2 from 5 September to 27 December 2026. Its spatial experience prompted the author to imagine twist not as a single cut or junction but as a relation distributed and accumulated through an entire space. This statement records the genealogy of the idea. The exhibition is not offered as physical evidence, and the mathematical and cosmological claims in this paper are the author's independent hypotheses.

2 The Finite State Motivation

WRRRA begins from an information ontology. The physical state is treated as finite and executable rather than as a completed actually infinite data object. In this paper that principle is an axiom of the proposed framework, not an empirical theorem about the universe.

Let the total physically available state be decomposed as

$$D_{total} = D_{rendered} + D_{residual}.$$

Visible matter is a phenotype produced by a rendering map,

$$M_{visible} = \Pi_M(D_{rendered}).$$

The central hypothesis is that a state need not be optically or materially rendered in order to remain dynamically present. Gravitational bookkeeping may therefore require

$$T_{\mu\nu}^{grav} = T_{\mu\nu}^{visible} + T_{\mu\nu}^T,$$

where $T_{\mu\nu}^T$ is not an invisible particle inventory by definition. It is the effective stress generated by residual twist information. The terms data, renderer, and phenotype refer to physical state ownership and transformation; they do not assert that the universe is a conventional digital computer.

The rejection of actual infinity motivates a finite model but does not determine whether the universe is spatially closed, temporally recurrent, or both. Those are additional model choices. The present proposal selects a compact boundaryless spatial sector with possible nontrivial holonomy as the minimal research target.

3 Finite Boundaryless and Accumulated Twist Closure

A finite interval has endpoints, but a circle is finite in length and has no endpoint. Likewise, a sphere is finite in area and has no boundary. In three dimensions a three sphere or a compact quotient such as a three torus can be finite and boundaryless. The phrase finite but boundaryless is therefore mathematically coherent without invoking a physical edge.

The Möbius strip is useful because its half twist is distributed around the whole band. No physical section is privileged as the place where the twist occurs. The important intuition is therefore not one cut followed by one identification, nor merely a single orientation flip after a circuit. It is that infinitesimal local twisting can accumulate into a finite global state. The strip is still not an admissible literal model of cosmic space because it is a two dimensional surface with a boundary. The cosmological hypothesis instead concerns a higher dimensional compact manifold or bundle with a distributed connection.

Let ω_T denote a local twist rate along a path. The accumulated open path twist is

$$\Theta_\gamma = \int_\gamma \omega_T.$$

For a connection A_T , closed transport produces the ordered holonomy

$$H(\gamma) = P \exp \left(\oint_\gamma A_T \right).$$

If transport repeats while the state evolves, the historical accumulation is

$$H_N(\gamma) = H_N H_{N-1} \cdots H_1.$$

In an Abelian approximation the corresponding twist phases add, $\Theta_N = \sum_{m=1}^N \Theta_m$. In a non Abelian model the order of the local increments matters. A static Möbius orientation flip is only a finite order special case and is too simple to represent the proposed accumulating cosmic state. The field $\Theta(t, x)$ denotes the effective local record of this accumulated transport. A nonidentity value $H(\gamma) \neq I$ means that a transported state returns with a residue, but the proposed physics resides in the distributed connection and its accumulated history rather than in an isolated end point identification.

Three distinctions are essential.

Concept	Question answered	What it does not imply
Compactness	Is the total spatial extent finite	Local curvature or stress
Curvature	Does local parallel transport depend on geometry	Nontrivial global identification
Holonomy	What accumulated transformation remains after closed transport	A unique stress tensor without dynamics

A flat three torus supplies the decisive counterexample. It is compact and boundaryless and can be locally Euclidean. Thus global closure alone cannot be identified with dark matter, dark energy, or curvature.

4 From Global Closure to Local Stress

The proposed causal structure has six distinct maps. None may be omitted if the theory is to carry explanatory content.

$$C \xrightarrow{F_A} A_T \xrightarrow{F_H} H \xrightarrow{F_R} \Theta \xrightarrow{F_S} (\rho_T, q_T) \xrightarrow{F_G} T_{\mu\nu}^T \xrightarrow{F_O} O.$$

Here C is the global closure class, A_T the distributed twist connection, H the accumulated holonomy data, Θ the local record of accumulated twist, (ρ_T, q_T) its magnitude and anisotropic response sectors, $T_{\mu\nu}^T$ the effective stress, and O the set of observables. The maps F_A through F_O are part of the physical model. Merely naming a topology does not provide them.

A candidate gravitational completion can be written as

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} (T_{\mu\nu}^{visible} + T_{\mu\nu}^T).$$

This equation is a bookkeeping target, not a derivation of $T_{\mu\nu}^T$. A complete theory must specify an action or constitutive law, derive the field equations, and satisfy

$$\nabla_\mu (T^{\mu\nu}_{visible} + T^{\mu\nu}_T) = 0.$$

The minimum effective decomposition relative to an observer field u^μ is

$$T^{\mu\nu}_T = \rho_T u^\mu u^\nu + p_T h^{\mu\nu} + 2u^{(\mu} Q_T^{\nu)} + \pi_T^{\mu\nu},$$

with $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$. The earlier galactic notation q_T is treated as a compact label for the anisotropic constitutive sector represented covariantly by Q_T^μ and $\pi_T^{\mu\nu}$. Their closure relations remain open.

5 Hybrid Expansion Twist Kinematics

Expansion and twist should not be collapsed into one word. A finite compact universe can expand, and a nontrivial global identification can evolve while the global volume changes little. The proposed kinematic separation is

$$g_{ij}(t, x) = a^2(t) \tilde{g}_{ij}(\Theta(t, x)).$$

The scale factor $a(t)$ controls isotropic volume scaling. The dimensionless state Θ controls shape, gluing, transport, or anisotropic deformation. This parameterization defines three nested cases.

Case	Expansion rate	Twist rate	Interpretation
Expansion only	$\dot{a} > 0$	$\dot{\Theta} = 0$	Standard geometric redshift sector
Pure twist	$\dot{a} = 0$	$\dot{\Theta} > 0$	Fixed global scale with evolving transport
Hybrid	$\dot{a} > 0$	$\dot{\Theta} > 0$	Both effects contribute

The pure twist limit is the most radical interpretation of the motivating idea. It is not assumed in advance. The hybrid model is retained because current observations establish an effective expansion history but do not by themselves decide whether every part of that history must be assigned to a single scale degree of freedom.

6 Photon Redshift

The exact general relativistic definition of observed redshift is

$$1 + z_{obs} = \frac{(u_\mu k^\mu)_e}{(u_\mu k^\mu)_o},$$

where k^μ is the photon wave vector and u^μ is the emitter or observer four velocity. In a homogeneous FLRW spacetime this reduces to $1 + z = a_0 / a_e$.

WRRRA proposes a conditional factorization

$$1 + z_{obs} = (1 + z_{exp})(1 + z_T) = \frac{a_0}{a_e} \exp(\Xi_T),$$

where Ξ_T is a line of sight transport functional derived from Θ and the photon path. A schematic local form is

$$\Xi_T[\gamma] = \int_{\gamma} K_T(\Theta, \nabla \Theta, \dot{\Theta}, k^\mu, u^\mu) d\lambda.$$

At low redshift the logarithmic rates approximately add,

$$H_{obs} \simeq H_{exp} + H_T.$$

This is not yet a prediction because the kernel K_T has not been derived from a frozen action. If $\Xi_T(z)$ is fitted independently for every distance indicator, the construction has no falsifying power. The decisive requirement is that the same Θ and the same kernel reproduce redshift observables while also fixing nonredshift phenomena.

7 One Twist State and Multiple Phenotypes

The proposed economy is expressed by a common state map.

State component	Calculated phenotype
$\dot{\Theta}$	Cosmic redshift contribution
$\nabla \Theta$	Additional galactic acceleration
$[n]$	Disk plane and warp geometry
ρ_T, q_T	Stress and vertical gravity

The same symbol does not guarantee the same physics. The model earns unification only when a single frozen state evolution and constitutive rule calculate all four sectors without independent retuning.

In WRRRA language, the chain is

$$\text{SOURCE} \rightarrow \text{renderer} \rightarrow \text{phenotype} \rightarrow \text{observable record}.$$

The SOURCE contains global topology, boundary data, state relations, and permitted evolution. The renderer maps the state into locally measurable fields. Visible matter is one phenotype. Additional gravitational response can be another. The phrase hidden data refers to a nonrendered physical state in this chain, not to a freely adjustable reservoir invoked after seeing the observations.

8 Galactic Weak Field Limit

The existing WRRRA galactic disk model defines a minimal twist state

$$T = ([n], \rho_T, q_T), n \sim -n.$$

The equivalence $n \sim -n$ makes $[n]$ an unoriented director. It selects a local plane without selecting an up direction or a rotation direction. The projector is

$$P_{ij} = \delta_{ij} - n_i n_j, P^2 = P.$$

A warped disk can be represented by the level surface

$$S(R, \phi, z, t) = z - z_T(R, \phi, t) = 0,$$

with

$$n = \frac{\nabla S}{\|\nabla S\|}.$$

This construction answers the user's original conceptual objection. A two dimensional disk inside three dimensional space need not begin with a global up and down. An unoriented plane field can select the local transport surface. Angular momentum remains a separate dynamical quantity generated by torques and conserved or redistributed within the selected plane. Twist selects the plane; it does not create rotation or determine its sign.

The magnitude sector is associated with an additional weak field acceleration. The prior model used the frozen scale

$$a_T = c H_0 \sqrt{\frac{f_T}{8}}$$

with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $f_T = 0.265$, obtaining

$$a_T = 1.1918 \times 10^{-10} \text{ m s}^{-2}.$$

With a frozen SPARC calibrated response and a Milky Way baryonic model, it returned circular speeds of 226.3 km s^{-1} at 8.2 kpc and 201.6 km s^{-1} at 25 kpc , with an RMS difference of 2.42 km s^{-1} from the selected linearized rotation curve over 8 to 25 kpc . The warp onset near 11 kpc and outer inclination near three degrees were Gaia calibrated rather than derived from first principles. These are postcalibration outputs of a quantitative toy model, not proof of the global hypothesis.

9 Relation to Existing Theories

Every empirical physical theory contains contact with observation. The relevant distinction is not whether calibration exists, but what is calibrated, what is then held fixed, and how many independent phenomena are subsequently computed.

Framework	Primary owner of the effect	Typical calibration	Strength	Present limitation relative to this question
FLRW with Lambda CDM	Scale factor plus matter and dark components	Cosmological parameters and initial spectrum	Joint fit to CMB BAO supernovae growth and lensing	Does not make galactic plane selection a consequence of global closure
Particle dark matter halos	Additional gravitating matter	Halo population and baryonic feedback relations	Flexible structure formation framework	Disk warps and angular momentum depend on environmental and baryonic histories
MOND and AQUAL	Modified low acceleration response	Universal acceleration scale and baryonic distributions	Economical galaxy regularities including RAR and BTFR	Relativistic cosmology lensing and cluster closure require completion
Tidal torque and disk formation	Torques from initial density field and angular momentum conservation	Initial conditions and baryonic processes	Established route to angular momentum acquisition	Does not by itself define one universal global plane or dark mass phenotype
WRRR twist model	Global transport residue rendered as stress and geometry	Global state law plus declared anchors	Attempts one owner for redshift residue extra gravity and plane geometry	Covariant action transport kernel and joint likelihood remain open

The WRRR proposal is not original merely because it uses different names. Its potentially distinctive content is the owner renderer phenotype assignment and the requirement that one frozen twist state support cosmological and galactic calculations. Conversely, if $a_{eff} = F(\Theta)$ reproduces every FLRW observable but Θ has no independently calculated consequence, the model is only an alternative interpretation or reparameterization.

10 Validation Architecture

Every major claim is evaluated with the same four stage chain.

Stage	Required content
Validation input	The observation constant boundary condition or external model used

Stage	Required content
WRRR specific transformation	The state ownership and frozen mathematical map
Output	A calculated value curve relation or qualitative structure
Falsification condition	A result that would reject or sharply weaken the map

Calibration is legitimate when it is declared. A reproduced known value can be an explanatory output if it follows through a WRRR specific structure. An unmeasured quantity calculated after the model is frozen is a prediction. An imported equation remains external support unless it is actually evaluated within the WRRR chain.

The same burden must be applied to alternatives. Lambda CDM is not rejected because it uses fitted parameters or simulations, and WRRR is not validated merely because standard models use them. Comparative assessment must count independent free functions, calibrated constants, held out data, and cross domain reuse.

11 Cosmological Closure Tests

11.1 Supernova Time Dilation

Observed Type Ia supernova light curves broaden approximately by $1 + z$. A twist redshift law must transform emission times and photon frequencies consistently, not merely shift spectral lines. Failure to reproduce the same temporal dilation falsifies a pure twist redshift mechanism.

11.2 Distance Duality and Tolman Dimming

For metric light propagation with photon number conservation,

$$d_L = (1 + z)^2 d_A.$$

The same null propagation law must also reproduce surface brightness dimming proportional to $(1 + z)^{-4}$. A kernel chosen only to match redshift but inconsistent with luminosity distance, angular diameter distance, or Tolman dimming is rejected.

11.3 Baryon Acoustic Oscillations

Radial BAO constrains an expansion related rate, while transverse BAO constrains an angular diameter distance. A hybrid theory must calculate both from the same metric and photon transport rule. It cannot assign radial BAO to twist and transverse BAO to expansion with unrelated functions.

11.4 Cosmic Microwave Background

The model must account for the near blackbody spectrum, the temperature redshift relation $T(z) = T_0(1+z)$ or a precisely derived alternative, acoustic angular scales, damping, polarization, and lensing. Existing searches have not detected a compact topology with a fundamental domain smaller than roughly the diameter of the last scattering surface. This constrains small compact models but does not prove that the universe is infinite.

11.5 Redshift Drift

The time derivative of source redshift provides a direct separation test because expansion and evolving transport need not predict the same drift. In the proposed factorization,

$$\frac{d}{dt_0} \ln(1+z_{obs}) = \frac{d}{dt_0} \ln\left(\frac{a_0}{a_e}\right) + \frac{d\Xi_T}{dt_0}.$$

A frozen Ξ_T evolution therefore gives a specific correction to the Sandage Loeb signal.

11.6 Growth and Lensing

The effective stress that changes galaxy dynamics must enter cosmological perturbations and light deflection consistently. The same metric potentials must confront weak lensing, redshift space distortions, cluster dynamics, and the matter power spectrum. Arbitrary gravitational slip introduced only after a lensing mismatch counts as a new model degree of freedom.

12 Galactic Closure Tests

The galactic sector supplies a nearer and more structured test bench.

First, one frozen acceleration scale and response must be evaluated across complete rotation curve samples, including residual correlations and uncertainty propagation. Second, the same stress state must calculate vertical velocity dispersion, disk thickness, and flaring through the vertical Jeans equation. Third, the same metric must reproduce gravitational lensing. Fourth, one evolving plane field should relate stellar, atomic gas, and molecular gas warp geometries and vertical motions. Fifth, Solar System constraints must be recovered without an object specific screening function added after the fact.

Polar rings, counterrotating disks, strongly interacting galaxies, and low surface brightness systems are especially informative because they separate plane selection, rotation direction, environmental torque, and additional gravity.

13 Frozen Model Protocol

A scientifically useful next version should freeze the following before opening a held out validation set.

1. The topology or global closure class C .
2. The distributed connection A_T and local twist rate ω_T .
3. The evolution equation for Θ .
4. The map from accumulated holonomy to local residue.
5. The constitutive map from Θ to $T_T^{\mu\nu}$.
6. The photon transport kernel K_T .
7. The weak field limit and plane director equation.
8. All universal constants and allowed nuisance parameters.
9. The data split and numerical failure thresholds.

The frozen model is then run in sequence rather than refitted independently to each sector. A proposed minimum objective is

$$\chi_{joint}^2 = \chi_{SN}^2 + \chi_{BAO}^2 + \chi_{CMB}^2 + \chi_{growth}^2 + \chi_{lens}^2 + \chi_{galaxy}^2 + \chi_{vertical}^2,$$

with complexity penalties applied to genuinely free parameters and functions. Passing only one term does not validate the common owner claim.

14 Claim Ledger

Claim	Grade	Present basis	What would change the grade
Actual infinity is not physically instantiated	Axiom and open	WRRR modeling principle	Independent physical necessity or contradiction
A universe can be finite and boundaryless	Exact mathematical possibility	Compact manifolds	No change required
A Möbius strip is the cosmic topology	Rejected literal claim	It has a boundary and wrong dimension	Not pursued
Möbius means one cut or one localized flip	Rejected interpretation	Its useful feature is distributed twist	Replace by a continuous connection
Accumulated holonomy may encode global twist	Structural hypothesis	Differential geometry permits it	Specify viable manifold bundle and connection
Topology alone generates gravity	Rejected inference	Flat compact counterexamples	Requires added dynamics
$g_{ij} = a^2 \tilde{g}_{ij}(\Theta)$	Exact kinematic parameterization	Variable separation	Dynamics must select a trajectory

Claim	Grade	Present basis	What would change the grade
Redshift factorizes into expansion and twist	Conditional hypothesis	Proposed photon transport form	Derive and fit a frozen kernel
Residual twist information gravitates	Causal hypothesis	WRRR owner assignment	Derive conserved stress and pass joint tests
Twist selects a disk plane but not angular momentum	Structural output	Unoriented director model	Test against multicomponent disk geometry
Milky Way rotation outputs	Postcalibration output	Frozen toy model	Reproduce with full uncertainties and held out data
Pure twist replaces expansion	Open limiting model	No completed likelihood	Pass all cosmological closure tests
Hybrid expansion and twist is admissible	Structural model space	Independent α and Θ	Constrain by joint observations
Covariant action lensing and vertical closure	Open	Not yet derived	Complete action and numerical implementation

15 Falsification Conditions

The common twist hypothesis is rejected or sharply weakened if any of the following occurs after the model is frozen.

1. The redshift kernel matches spectra but fails supernova time dilation, distance duality, Tolman dimming, BAO, or CMB temperature scaling.
2. A single Θ evolution cannot fit cosmological distances and redshift drift without a new function for each probe.
3. The stress needed for rotation curves fails gravitational lensing, structure growth, clusters, or Solar System constraints.
4. The plane director required for warps is unrelated to the stress magnitude required for additional gravity.
5. Galaxy by galaxy values of a_T or unrelated plane fields are required after the universal state is declared.
6. No conserved and stable covariant completion can realize the proposed effective stress.
7. The model reduces exactly to FLRW through $a_{eff} = F(\Theta)$ and produces no independently calculated non FLRW observable.

These conditions are intentionally stronger than reproducing one rotation curve. The theory claims a common owner, so it must survive common ownership tests.

16 Research Program

The shortest path from structural hypothesis to physical theory has four stages.

Stage one Mathematical closure Select a compact boundaryless manifold or bundle, define a distributed connection with no privileged seam, calculate pathwise and repeated holonomy, and determine which accumulated degrees of freedom can survive locally.

Stage two Dynamical closure Propose an action $S[g, \Theta, \psi]$, derive the metric and twist equations, verify conservation, stability, causality, and the absence of uncontrolled ghosts or gradient instabilities.

Stage three Numerical closure Implement homogeneous background and linear perturbations, then compute supernova, BAO, CMB, growth, and lensing observables. Derive the weak field and disk equations from the same parameter set.

Stage four Preregistered validation Freeze parameters on a declared training set, publish hashes and thresholds, and score held out cosmological and galactic data. The result should be reported even if negative.

17 Conclusion

The motivating idea can be expressed rigorously without claiming more than has been shown. A finite universe does not need an edge. A Möbius strip provides a minimal picture of twist distributed around a closed form, but it is not the universe and it does not imply a privileged cosmic cut. Finiteness does not imply twist, and twist does not imply gravity. WRRR therefore adds a testable chain from global closure through a distributed connection, accumulated holonomy, and transport residue to effective stress and observables.

The proposal's most important move is to separate expansion from twist. The universe may expand, its twist may deepen, or both may occur. The observed redshift is then a product rather than an informal sum, while low redshift logarithmic rates may add. The model becomes distinct from standard cosmology only when the same frozen twist state also calculates additional gravity, disk plane geometry, warp, vertical response, lensing, and growth.

Visible mass is interpreted as a rendered phenotype, while residual state may remain gravitationally active. That interpretation is currently a causal hypothesis. The existing Milky Way calculation shows that a signless plane field and a stress magnitude sector can be kept conceptually separate from angular momentum and can be placed in one auditable toy model. The next decisive step is a covariant action and a joint cosmological galactic likelihood. Until then, the correct status is a structured, quantitative, and falsifiable research program.

Appendix A Notation

Symbol	Meaning
$a(t)$	Isotropic cosmological scale factor
Θ	Global and local twist state
ω_T	Local infinitesimal twist rate
A_T	Distributed twist connection
$H(\gamma)$	Holonomy around closed path γ
ρ_T	Twist stress energy density sector
q_T	Compact galactic notation for anisotropic response
$T_T^{\mu\nu}$	Effective twist stress tensor
$[n]$	Unoriented disk plane director with $n \sim -n$
$\mathcal{E}_T$	Integrated twist contribution to photon redshift
a_T	Frozen weak field transition acceleration
$D_{residual}$	Nonrendered but dynamically available state

Appendix B Preregistered Test Matrix

Domain	Frozen input	WRR transformation	Held out output	Failure rule
Supernovae	Calibration subset and light curve model	Common photon kernel	Distance and time dilation	Inconsistent $1+z$ scaling
BAO	Sound horizon treatment	Same background metric	Radial and transverse distances	Separate ad hoc corrections required
CMB	Initial state and recombination physics	Background plus perturbations	Peaks damping polarization lensing	Joint likelihood fails threshold
Redshift drift	Frozen a and Θ history	Time derivative of factorized redshift	Sign and amplitude by redshift	Drift differs beyond preregistered error
Galaxies	Baryonic maps and universal constants	One weak field law	Rotation curves RAR BTFR	Object specific law required

Domain	Frozen input	WRR transformation	Held out output	Failure rule
Disk structure	Same galactic state	Director and vertical stress	Warp flaring dispersion	Geometry and stress decouple
Lensing	Same metric stress	Null geodesics	Galaxy and cluster lensing	New independent lensing source required
Solar System	Same universal law	High acceleration limit	Ephemeris compatible residuals	Screening added only for local tests

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Related WRRRA Records

- WRRRA Core <https://doi.org/10.5281/zenodo.22650956>
- Twist Stress Universe and the Mass Phenotype of Dark Matter
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- WRRRA Spatial Twist Selection of Galactic Disk Planes and the Dark Mass Phenotype
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